

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI, RAJ.
I Semester 2026-27 CS/ECE/EEE/INSTR F215 Digital Design

Max Marks: 15

13 Sept 2026

Quiz 2

Max Duration: 30 minutes

ID No:	Name:
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Instructions: Only write **final answers** in the boxes provided. **No other writing on the sheet.** Do not overwrite.

Q1. The truth table for a function $F(A,B,C)$ was recorded in the order the tests happened to run, as follows. Complete the K-map on the right using this data, then find the minimal SOP.

Truth table:

Fill in the K-map (1 marks):

A	B	C	F
0	1	1	1
1	1	0	1
0	0	0	1
1	0	1	0
0	0	1	1
1	1	1	0
0	1	0	1
1	0	0	0

A\BC	00	01	11	10
0	1	1	1	1
1	0	0	0	1

All or none, no partial marking

Q2. Find the minimal SOP for the function $F(A,B,C,D)$ given by the K-map below. List the prime implicants in the boxes on the right, ticking the box under each one that is essential, then give the minimal SOP.

K-map:

Prime implicants (3 marks):

AB\CD	00	01	11	10
00	0	0	X	1
01	1	0	1	1
11	1	0	0	X
10	0	0	1	0

BD'	EPI? ☒
A'C	EPI? ☒
B'CD	EPI? ☒
	EPI? ☐

0.5 for each correct term
 0.5 for marking EPI or not
 -1 per term for extra terms (min
 ≥ 0 for the entire question)

Give marks if students have
 shown labeled boxes without
 writing the algebraic expressions
 IN REASONABLE CASES.

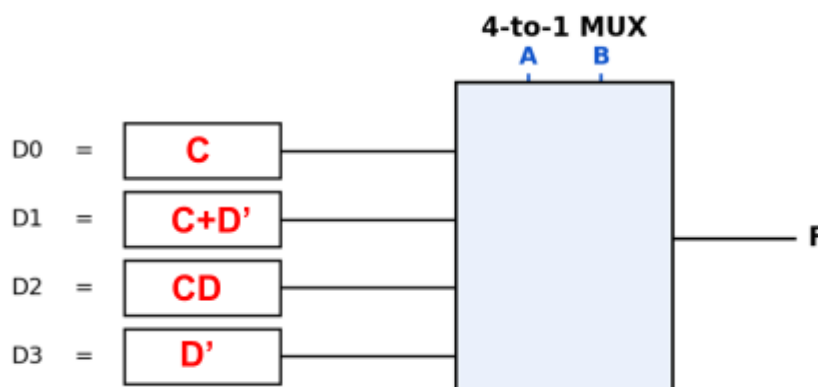
MSP (1 mark):

$BD' + A'C + B'CD$

all or none

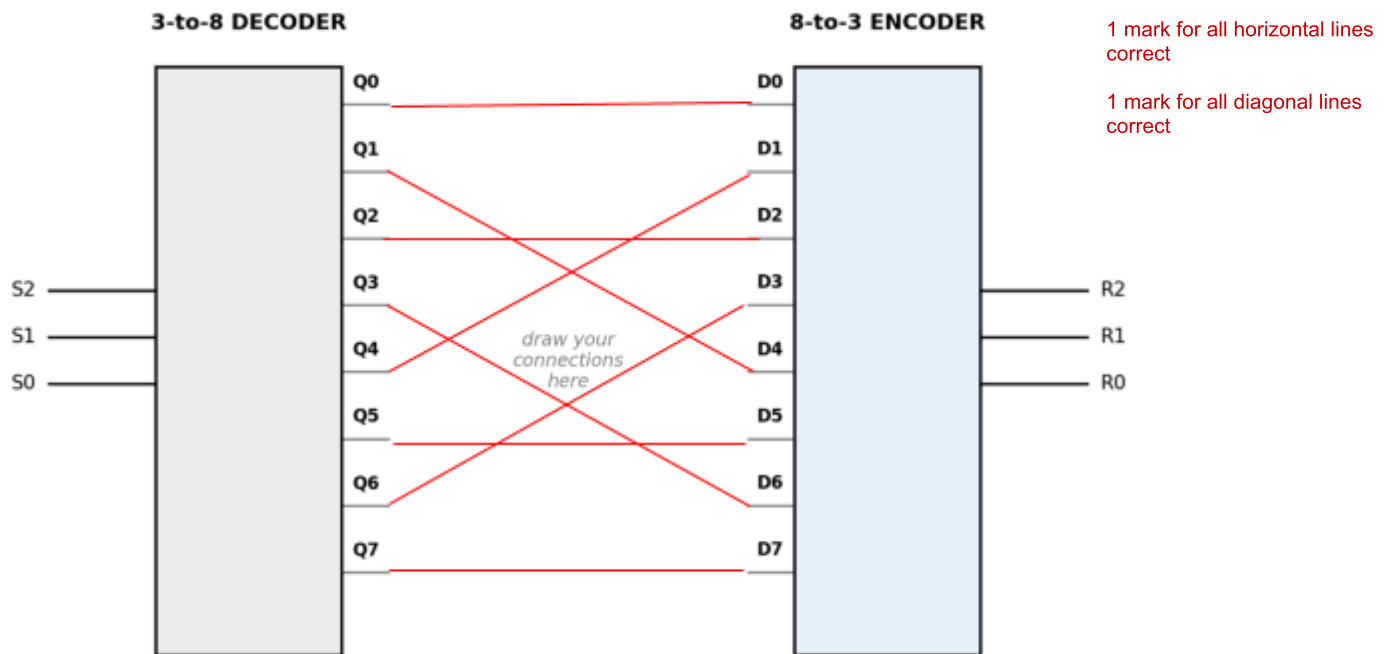
If students have used different variable names (WXYZ instead of ABCD), ok to give marks as long as it is reasonable. Some will use random names like XWYZ etc. Those you can give by your discretion.

Q3. Implement the same function $F(A,B,C,D)$ from Q2 using the 4-to-1 multiplexer shown below, with A and B as the select (switching) inputs. Fill in each data input box with the correct minimal expression (only true and complement forms of literals are available). [4 marks]



1 mark for each line, no partial per line

Q4. A sensor transmits its 3-bit reading serially, LSB-first. The receiving circuit, however, was wired assuming the bits would arrive MSB-first — so every bit of the received pattern ends up reversed from what was actually sent (110 becomes 011). The following circuit is used to correct the received bit pattern on the receiver side (eg. the received pattern 011 is converted back to 110), using only one 3-to-8 decoder and one 8-to-3 encoder. Complete the connections below. **[2 marks]**



Q5. A step counter updates its step count every hour by adding the latest hour's steps to the previous total, using a single binary adder. Both values are stored in units of hundreds of steps (e.g. a stored value of 20 represents 2000 steps), and the adder displays the new total, assuming it is an unsigned integer. The engineering team collected the following test log (all values in hundreds of steps):

Previous total	Latest steps	New total (displayed)	Wrong?
20	15	35	
50	13	63	
50	14	0	☒
45	25	6	☒
10	5	15	

1 mark all or none for ticking all 'wrong' rows

Some of these results are wrong.

1. Identify which (tick the erroneous rows) **[1 mark]**

2. Determine the exact bit-width of the adder used **[3 marks]**: 6

3 marks all or none for giving the final answer in an unambiguous manner.

Partial as per understanding if required (instructor's discretion)

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Q1. The truth table for a function $F(A,B,C)$ was recorded in the order the tests happened to run, as follows. Complete the K-map on the right using this data, then find the minimal SOP.

Truth table:

Fill in the K-map (1 marks):

A	B	C	F
0	1	1	1
1	1	0	0
0	0	0	1
1	0	1	0
0	0	1	1
1	1	1	1
0	1	0	1
1	0	0	0

A\BC	00	01	11	10
0	1	1	1	1
1	0	0	1	0

Q2. Find the minimal SOP for the function $F(A,B,C,D)$ given by the K-map below. List the prime implicants in the boxes on the right, ticking the box under each one that is essential, then give the minimal SOP.

K-map:

Prime implicants (3 marks):

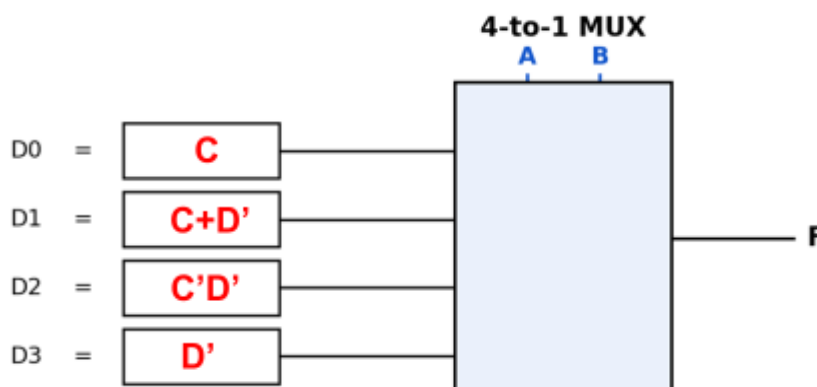
AB\CD	00	01	11	10
00	0	0	X	1
01	1	0	1	1
11	1	0	0	X
10	1	0	0	0

BD'	EPI? ☒
A'C	EPI? ☒
AC'D'	EPI? ☒
	EPI? ☐

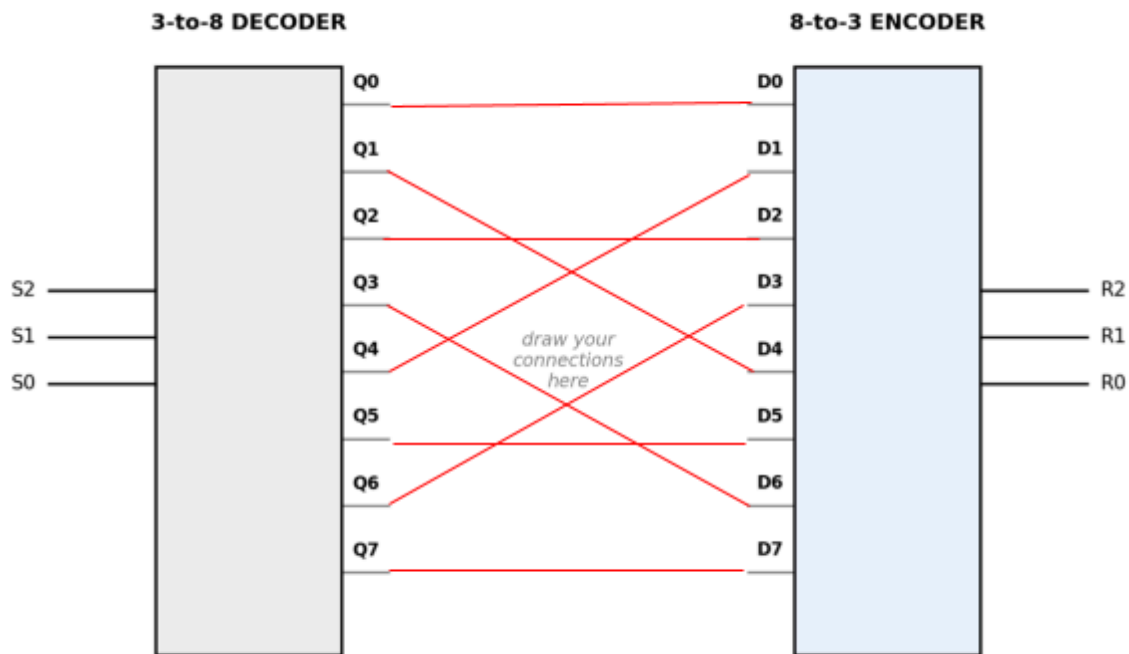
MSP (1 mark):

$BD' + A'C + AC'D'$

Q3. Implement the same function $F(A,B,C,D)$ from Q2 using the 4-to-1 multiplexer shown below, with A and B as the select (switching) inputs. Fill in each data input box with the correct minimal expression (only true and complement forms of literals are available). [4 marks]



Q4. A sensor transmits its 3-bit reading serially, LSB-first. The receiving circuit, however, was wired assuming the bits would arrive MSB-first — so every bit of the received pattern ends up reversed from what was actually sent (110 becomes 011). The following circuit is used to correct the received bit pattern on the receiver side (eg. the received pattern 011 is converted back to 110), using only one 3-to-8 decoder and one 8-to-3 encoder. Complete the connections below. **[2 marks]**



Q5. A step counter updates its step count every hour by adding the latest hour's steps to the previous total, using a single binary adder. Both values are stored in units of hundreds of steps (e.g. a stored value of 20 represents 2000 steps), and the adder displays the new total, assuming it is an unsigned integer. The engineering team collected the following test log (all values in hundreds of steps):

Previous total	Latest steps	New total (displayed)	Wrong?
20	12	0	☒
20	10	30	☐
14	14	28	☐
15	25	8	☒
10	5	15	☐

Some of these results are wrong.

1. Identify which (tick the erroneous rows) **[1 mark]**
2. Determine the exact bit-width of the adder used **[3 marks]**: 5